

equation has $(n-r)$ linearly independent solutions. Since a linearly independent system can never be zero, therefore, the given system must have a non-zero solution.

EXAMPLE 13.61

Solve

$$x + 3y - 2z = 0$$

$$2x - y + 4z = 0$$

$$x - 11y + 14z = 0.$$

Solution. The matrix form of the given system of homogeneous equations is $AX = 0$, where

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

We note that

$$|A| = \begin{vmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{vmatrix} = 30 - 72 + 42 = 0.$$

Therefore A is singular, that is $\rho(A) < n$. Thus, the given system has a non-trivial solution and will have infinite number of solutions.

The given system is

$$\begin{aligned} & \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \\ & \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0, \quad \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \\ & \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0, \quad R_3 \rightarrow R_3 - 2R_2 \end{aligned}$$

and so we have

$$x + 3y - 2z = 0$$

$$-7y + 8z = 0.$$

These equations yield $y = \frac{8}{7}z, x = \frac{-10}{7}z$. Giving different values to z , we get infinite number of solutions.

EXAMPLE 13.62

Solve

$$x_1 - x_2 + x_3 = 0$$

$$x_1 + 2x_2 - x_3 = 0$$

$$2x_1 + x_2 + 3x_3 = 0.$$

Solution. In matrix form, we have $AX = 0$, where

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

We note that $|A| = 9 \neq 0$. Thus A is non-singular. Hence, the given system of homogeneous equation has only trivial solution $x_1 = x_2 = x_3 = 0$.

EXAMPLE 13.63

Solve

$$2x - 2y + 5z + 3w = 0$$

$$4x - y + z + w = 0$$

$$3x - 2y + 3z + 4w = 0$$

$$x - 3y + 7z + 6w = 0.$$

Solution. The matrix form of the given system of homogeneous equation is

$$AX = \begin{bmatrix} 2 & -2 & 5 & 3 \\ 4 & -1 & 1 & 1 \\ 3 & -2 & 3 & 4 \\ 1 & -3 & 7 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = 0.$$

Performing row elementary transformations to get echelon form of A , we have

$$\begin{aligned} A &= \begin{bmatrix} 2 & -2 & 5 & 3 \\ 4 & -1 & 1 & 1 \\ 3 & -2 & 3 & 4 \\ 1 & -3 & 7 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 7 & 6 \\ 4 & -1 & 1 & 1 \\ 3 & -2 & 3 & 4 \\ 2 & -2 & 5 & 3 \end{bmatrix} \quad R_1 \leftrightarrow R_4 \\ &\sim \begin{bmatrix} 1 & -3 & 7 & 6 \\ 4 & 11 & -27 & -23 \\ 0 & 7 & -18 & -14 \\ 0 & 4 & -9 & -9 \end{bmatrix} \quad \begin{matrix} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{matrix} \\ &\sim \begin{bmatrix} 1 & -3 & 7 & 6 \\ 0 & 4 & -9 & -9 \\ 0 & 7 & -18 & -14 \\ 0 & 4 & -9 & -9 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_3 \\ &\sim \begin{bmatrix} 1 & -3 & 7 & 6 \\ 0 & 4 & -9 & -9 \\ 0 & 28 & -72 & -56 \\ 0 & 4 & -9 & -9 \end{bmatrix} \quad \begin{matrix} R_3 \rightarrow 4R_3 \\ R_4 \rightarrow R_4 - R_2 \end{matrix} \\ &\sim \begin{bmatrix} 1 & -3 & 7 & 6 \\ 0 & 4 & -9 & -9 \\ 0 & 0 & -9 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} R_3 \rightarrow R_3 - 7R_2 \\ R_4 \rightarrow R_4 - R_2 \end{matrix} \end{aligned}$$

The above echelon form of A suggests that rank of A is equal to the number of non-zero rows. Thus $\rho(A) = 3$.

The number of unknowns is 4. Thus $\rho(A) < n$. Hence, the given system possesses non-trivial solution. The number of independent solution will be $(n - r) = 4 - 3 = 1$.

Further, the given system is equivalent to

$$\begin{bmatrix} 1 & -3 & 7 & 6 \\ 0 & 4 & -9 & -9 \\ 0 & 0 & -9 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = 0$$

and so, we have

$$x - 3y + 7z + 6w = 0$$

$$4y - 9z - 9w = 0$$

$$-9z + 7w = 0$$

These equations yield $z = \frac{7}{9}w$, $y = 4w$, $x = \frac{5}{9}w$. Thus taking $w = t$, we get $x = \frac{5}{9}t$, $y = 4t$, $z = \frac{7}{9}t$, $w = t$ as the general solution of the given equations.

EXAMPLE 13.64

Determine the value of ρ For which the following equations have non-zero solutions:

$$x + 2y + 3z = \lambda x$$

$$3x + y + 2z = \lambda y$$

$$2x + 3y + z = \lambda z.$$

Solution. The matrix form of the given equation is

$$AX = \begin{bmatrix} 1 - \lambda & 2 & 3 \\ 3 & 1 - \lambda & 2 \\ 2 & 3 & 1 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

The given system will have non-zero solution only if $|A| = 0$, that is, if rank of A is less than 3. Thus for the existence of non-zero solution, we must have

$$\begin{vmatrix} 1 - \lambda & 2 & 3 \\ 3 & 1 - \lambda & 2 \\ 2 & 3 & 1 - \lambda \end{vmatrix} = 0$$

or

$$\begin{vmatrix} 6 - \lambda & 6 - \lambda & 6 - \lambda \\ 3 & 1 - \lambda & 2 \\ 2 & 3 & 1 - \lambda \end{vmatrix} = 0 \text{ using } R_1 \rightarrow R_1 + R_2 + R_3$$

or

$$(6 - \lambda) \begin{vmatrix} 1 & 1 & 1 \\ 3 & 1 - \lambda & 2 \\ 2 & 3 & 1 - \lambda \end{vmatrix} = 0$$

or

$$(6 - \lambda) \begin{vmatrix} 1 & 0 & 0 \\ 3 & -2 - \lambda & -1 \\ 2 & 1 & -1 - \lambda \end{vmatrix} = 0, \quad \begin{matrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{matrix}$$

or

$$(6 - \lambda)[\lambda^2 + 3\lambda + 3] = 0,$$

which yields

$$\lambda = 6 \text{ and } \frac{-3 \pm \sqrt{9 - 12}}{2}.$$

Thus, the only real value of λ for which the given system of equation has a solution is 6.

13.29 CHARACTERISTIC ROOTS AND VECTORS

Let A be a matrix of order n , λ a scalar and $X =$

$$\begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \text{ a column vector.}$$

Consider the equation

$$AX = \lambda X \quad (29)$$

Clearly $X = 0$ is a solution of (29) for any value of λ . The question arises whether there exist scalar λ and non-zero vector X , which simultaneously satisfy the equation (29). This problem is known as *characteristic value problem*. If I_n is unit matrix of order n , then (29) may be written in the form

$$(A - \lambda I_n)X = 0. \quad (30)$$

The equation (30) is the matrix form of a system of n homogeneous linear equations in n unknowns. This system will have a non-trivial solution if and only if the determinant of the coefficient matrix $A - \lambda I_n$ vanishes, that is, if

$$|A - \lambda I_n| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix} = 0.$$

The expansion of this determinant yields a polynomial of degree n in λ , which is called the *characteristic polynomial* of the matrix A .

The equation $|A - \lambda I_n| = 0$ is called the *characteristic equation* or *secular equation* of A .

The n roots of the characteristic equation of a matrix A of an order n are called the *characteristic roots*, *characteristic values*, *proper values*, *eigen values*, or *latent roots* of the matrix A .

EXAMPLE 13.76

Diagonalize the matrix

$$A = \begin{bmatrix} 2 & 1-2i \\ 1+2i & -2 \end{bmatrix}.$$

Solution. The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1-2i \\ 1+2i & -2-\lambda \end{vmatrix} = 0$$

or
$$\lambda^2 - 9 = 0,$$

which yields the characteristic roots as $\lambda = -3, 3$.
The eigenvectors corresponding to the eigenvalue -3 is given by $(A + 3I)X = 0$, that is, by

$$\begin{bmatrix} 5 & 1-2i \\ 1+2i & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which yields

$$5x_1 - (1-2i)x_2 = 0$$

$$(1+2i)x_2 + x_2 = 0.$$

Solving these equations, we get $x_1 = 1-2i, x_2 = -5$.

Hence, $X_1 = \begin{bmatrix} 1-2i \\ -5 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = -3$.

The eigenvector corresponding to $\lambda = 3$ is given by $(A - 3I)X = 0$ that is, by

$$\begin{bmatrix} -1 & 1-2i \\ 1+2i & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which yields

$$-x_1 + (1-2i)x_2 = 0$$

$$(1+2i)x_1 - 5x_2 = 0.$$

Solving these equations, we get $x_1 = 5, x_2 = 1+2i$.

Thus the required eigenvector is $X_2 = \begin{bmatrix} 5 \\ 1+2i \end{bmatrix}$.

We note that

$$X_2^H X_1 = [5 \quad 1-2i] \begin{bmatrix} 1-2i \\ -5 \end{bmatrix}$$

$$= 5(1-2i) - 5(1-2i) = 0.$$

Thus $\{X_1, X_2\}$ is an orthogonal set. Now

$$\text{Norm of } X_1 = \sqrt{|1-2i|^2 + |-5|^2} = \sqrt{5+25}$$

$$= \sqrt{30}$$

$$\text{Norm of } X_2 = \sqrt{|5|^2 + |1+2i|^2}$$

$$= \sqrt{25+5}$$

$$= \sqrt{30}.$$

Therefore, normalized characteristic vectors are

$$U_1 = \begin{bmatrix} \frac{1-2i}{\sqrt{30}} \\ \frac{-5}{\sqrt{30}} \end{bmatrix}, \quad U_2 = \begin{bmatrix} \frac{5}{\sqrt{30}} \\ \frac{1+2i}{\sqrt{30}} \end{bmatrix}.$$

Hence the transforming unitary matrix is

$$U = \begin{bmatrix} \frac{1-2i}{\sqrt{30}} & \frac{5}{\sqrt{30}} \\ \frac{-5}{\sqrt{30}} & \frac{1+2i}{\sqrt{30}} \end{bmatrix}.$$

We then note that

$$U^H A U = U^{-1} A U = \begin{bmatrix} -3 & 0 \\ 0 & 3 \end{bmatrix} = \text{diag} [-3 \ 3].$$

EXAMPLE 13.77

Diagonalize the Hermitian matrix

$$A = \begin{bmatrix} 5 & 2 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 5 & -2 \\ 0 & 0 & -2 & 2 \end{bmatrix}.$$

Solution. The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 5-\lambda & 2 & 0 & 0 \\ 2 & 2-\lambda & 0 & 0 \\ 0 & 0 & 5-\lambda & -2 \\ 0 & 0 & -2 & 2-\lambda \end{vmatrix} = 0.$$

The characteristic roots are 1, 1, 6, 6. The characteristic vector corresponding to $\lambda = 1$ are given by $(A - I)X = 0$, that is, by

$$\begin{bmatrix} 4 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 4 & -2 \\ 0 & 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

which yields

$$4x_1 + 2x_2 = 0$$

$$2x_1 + x_2 = 0$$

$$4x_3 - 2x_4 = 0$$

$$2x_3 + x_4 = 0$$

with the complete solution as

$$X_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}.$$

These vectors are already orthogonal. The normalized vectors are

$$U_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \\ 0 \\ 0 \end{bmatrix} \text{ and } U_2 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}.$$

Similarly, the normalized vectors corresponding to $\lambda = 6$ are

$$U_3 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \\ 0 \end{bmatrix} \text{ and } U_4 = \begin{bmatrix} 0 \\ 0 \\ -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}.$$

Hence, the transforming unitary matrix is

$$U = \begin{bmatrix} \frac{1}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} & 0 \\ -\frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} & 0 \\ 0 & \frac{1}{\sqrt{5}} & 0 & -\frac{2}{\sqrt{5}} \\ 0 & \frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \end{bmatrix} = 0.$$

and $U^0 AU = \text{diag} [1 \ 1 \ 6 \ 6]$.

13.35 TRIANGULARIZATION OF AN ARBITRARY MATRIX

Not every matrix can be reduced to diagonal form by a unitary transformation. But it is always possible to reduce a square matrix to a triangular form. In this direction, we have the following result.

Theorem 13.79. (Jacobi-Thoerem). Every square matrix A over the complex field can be reduced by a unitary transformation to upper triangular form with the characteristic roots on the diagonal.

Proof: We shall prove the theorem by induction on the order n of the matrix A . If $n = 1$, the theorem is obviously true. Suppose that the result holds for all matrices of order $n - 1$. Let λ_1 be the characteristic root of A and U_1 denote the corresponding unit characteristic vector. Then $AU_1 = \lambda_1 U_1$. Let $\{U_1, U_2, \dots, U_n\}$ be an orthonormal set, that is, $U = [U_1, U_2, \dots, U_n]$. Then

$$\begin{aligned} U^0 AU &= \begin{bmatrix} U_1^0 \\ U_2^0 \\ \dots \\ U_n^0 \end{bmatrix} [AU_1 \ AU_2 \ \dots \ AU_n] \\ &= \begin{bmatrix} U_1^0 \\ U_2^0 \\ \dots \\ U_n^0 \end{bmatrix} [\lambda_1 U_1 \ AU_2 \ \dots \ AU_n] \end{aligned}$$

Since $U_1^0 U_1 = I$ and $U_2^0 U_1 = U_3^0 U_1 = \dots = U_n^0 U_1 = 0$, we have

$$\begin{aligned} U^* AU &= \begin{bmatrix} \lambda_1 & U_1^0 AU_2 & \dots & \dots & U_1^0 AU_n \\ 0 & U_2^0 AU_2 & \dots & \dots & U_2^0 AU_n \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & U_n^0 AU_2 & \dots & \dots & U_n^0 AU_n \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 & B \\ 0 & C \end{bmatrix}. \end{aligned}$$

Now, by induction hypothesis, the matrix

$$C = \begin{bmatrix} U_2^0 AU_2 & \dots & \dots & U_2^0 AU_n \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ U_n^0 AU_2 & \dots & \dots & U_n^0 AU_n \end{bmatrix},$$

which is of order $n - 1$, is triangularizable. Thus there exists a unitary matrix W of order $n - 1$ which triangularize C , that is, $W^0 CW$ is triangular. Let

$$V = \begin{bmatrix} I & 0 \\ 0 & W \end{bmatrix}. \text{ Then } V^{-1} = \begin{bmatrix} I & 0 \\ 0 & W^{-1} \end{bmatrix} \text{ and}$$

$$V^0 = \begin{bmatrix} I & 0 \\ 0 & W^0 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & W^{-1} \end{bmatrix} = V^{-1}.$$

Hence V is unitary and

$$\begin{aligned} V^0 (U^0 AU) V &= \begin{bmatrix} I & 0 \\ 0 & W^0 \end{bmatrix} \begin{bmatrix} \lambda_1 & B \\ 0 & C \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & W \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 & B \\ 0 & W^0 C \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & W \end{bmatrix} = \begin{bmatrix} \lambda_1 & BW \\ 0 & W^0 CW \end{bmatrix}, \end{aligned}$$

where $W^0 CW$ is upper triangular. Thus, we have

$$\begin{aligned} (UV)^0 A (UV) &= \begin{bmatrix} \lambda_1 & BW \\ 0 & W^0 CW \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 & b_{12} & \dots & \dots & b_{1n} \\ 0 & \lambda_2 & \dots & \dots & b_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \dots & \lambda_n \end{bmatrix}. \end{aligned}$$

Since UV is unitary, the characteristic roots of the triangular matrix are the same as that of A . Thus, diagonal elements of triangular matrix are characteristic roots of A .

Theorem 13.80. A matrix A over the complex field can be diagonalized by a unitary transformation if and only if A is normal.

Proof: Suppose first that U is unitary and A can be diagonalized, that is, $U^0 AU = \text{diag}[\lambda_1 \ \lambda_2 \ \dots \ \lambda_n]$.